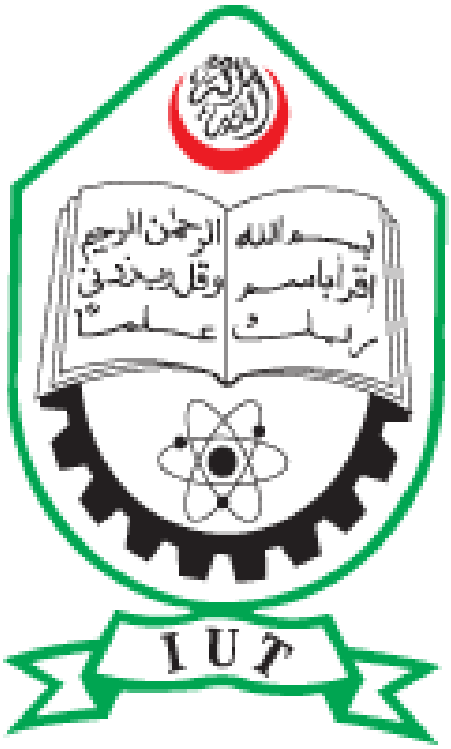


ISLAMIC UNIVERSITY OF TECHNOLOGY

SAP-1 Architecture Based 8-Bit Computer Design & Implementation

Team: Silicon 6

Course Code: EEE 4308

Course Name: Digital Electronics Lab

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Phase A: Clock Generator & Program Counter

1.1 Clock Pulse Generator

Overview

The clock signal is generated using a 555 timer IC with two modes of operation: Auto and Manual. Both modes are connected via an OR-gate to select the desired clock pulse signal.

Modes of Operation:

1. **Auto Mode (Astable Multivibrator):** Free-running oscillator producing a continuous square-wave signal. Duty cycle is approximately 48% with a time period of 0.33s.
2. **Manual Mode (Monostable Multivibrator):** Requires an external trigger (push button). Each press generates one clock pulse, allowing manual advancement of the instruction cycle.
3. **Halt Control:** The clock pulse input is ANDed with the NOT of the HALT input. When HALT is active (Logic 1), the clock generator input becomes zero, halting operation.

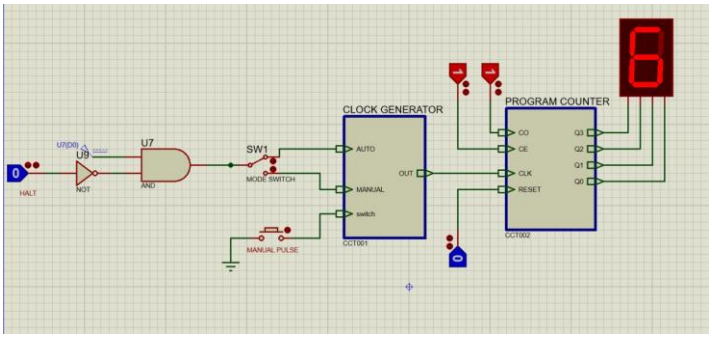


Fig – Circuit for Phase A

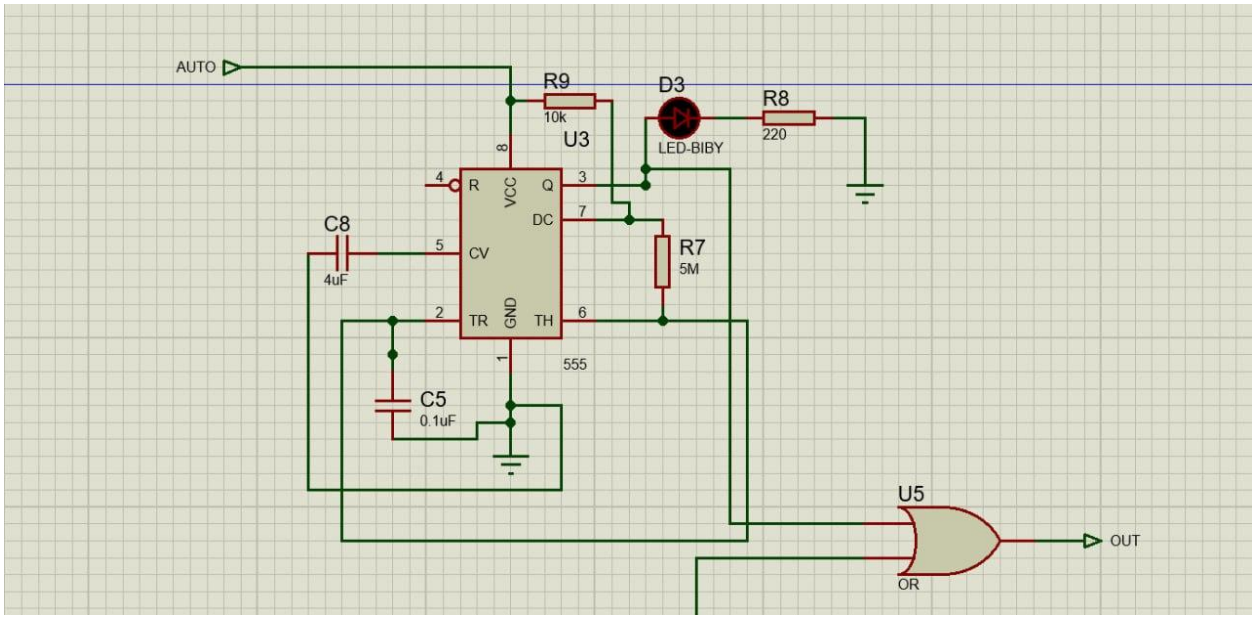


Fig- Astable multivibrator for auto mode

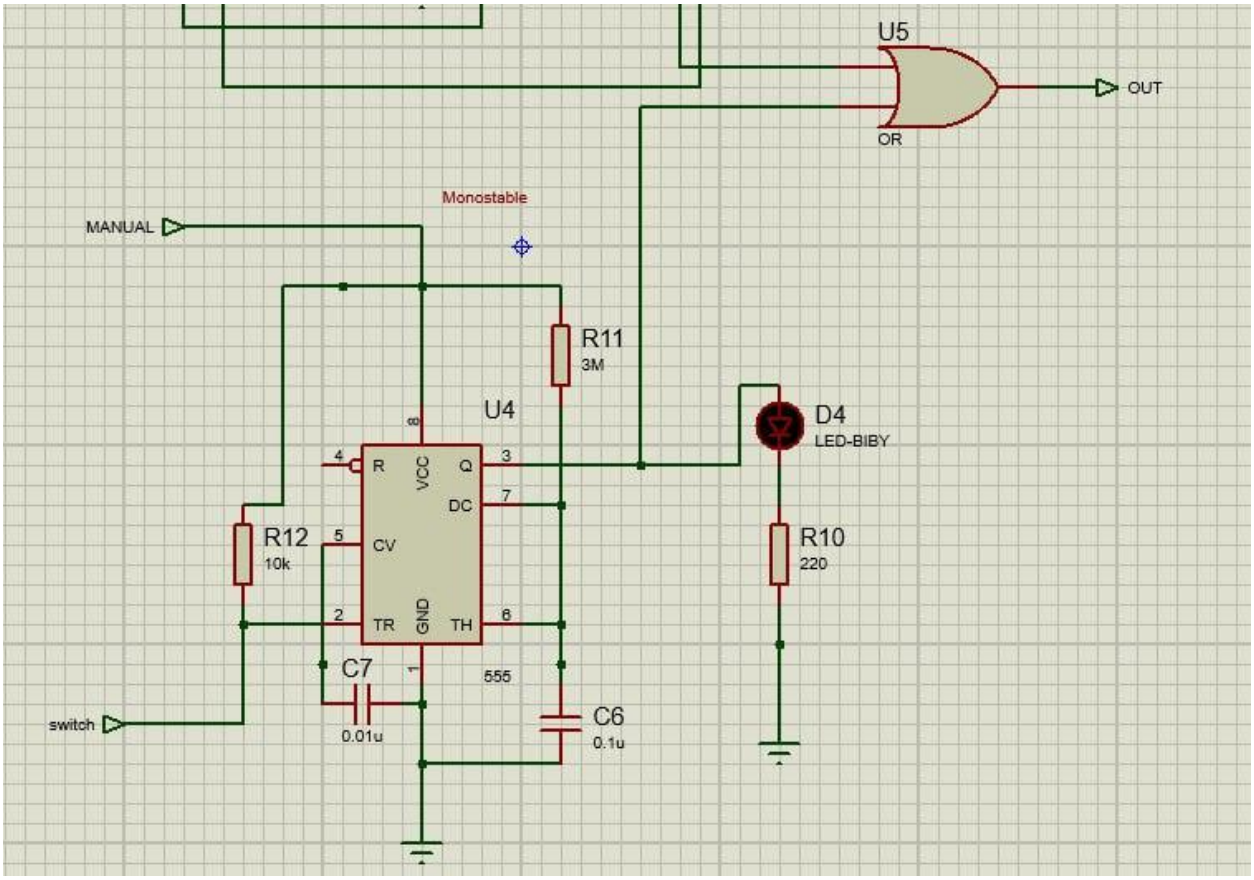


Fig- Monostable Multivibrator for manual mode

1.2 Program Counter Module

Overview

The Program Counter (PC) is a 4-bit register capable of counting from 0000 to 1111. It stores and provides the memory address of the next instruction. After each instruction cycle, the PC automatically increments.

Pin Configuration

Pin	Type	Function
CE	Input	Program Counter Enable: Increments PC on next clock cycle.
CO	Input	Program Counter Out: Places current value onto the bus.
RESET	Input	Resets the program counter to 0000.
CLK	Input	Clock pulse input.
OUT	Out	4-bit counter output.

Working Principle

The circuit consists of 4 D flip-flops with input D connected to output Q'. This configuration allows the flip-flop to toggle with each clock pulse. A 3-bit counter counts to 6 to match the 6-clock pulse instruction cycle. The output is connected to a tri-state buffer to ensure no output is given to the bus when CO is low.

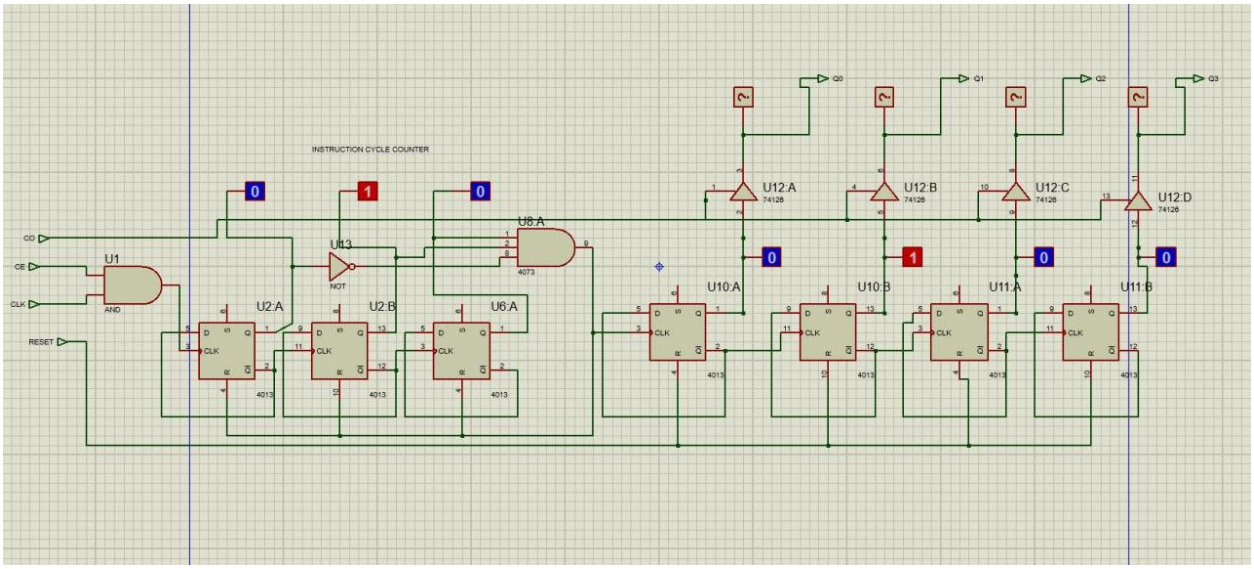


Fig: Program counter

Phase B: Registers & Memory System

2.1 Registers (Accumulator, B-Register, MAR)

Overview

Three primary registers were implemented using D flip-flops (IC 4013) and tri-state buffers. They store data and control transfer on the common bus.

1. Accumulator (A-Register)

Primary working register storing arithmetic results.

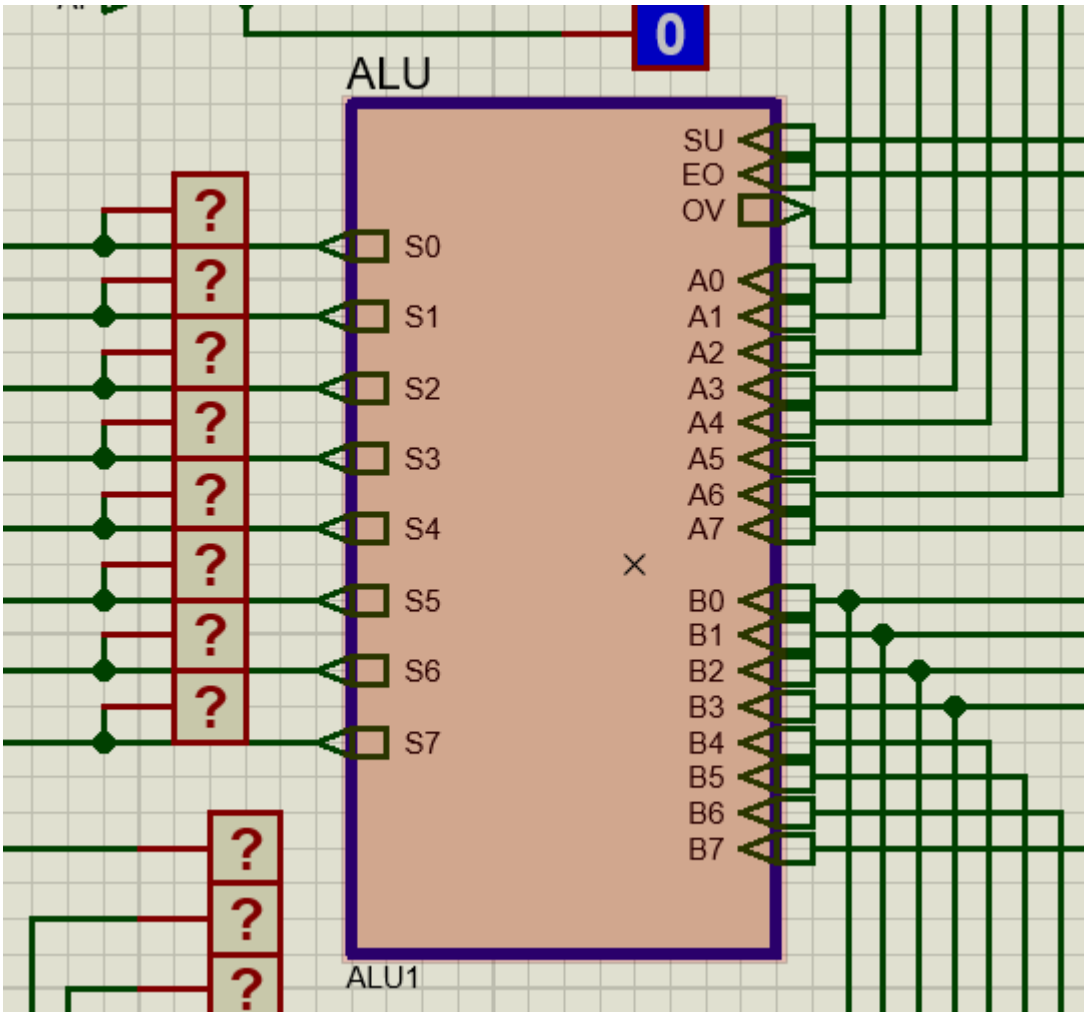


Fig – Accumulator Block Diagram

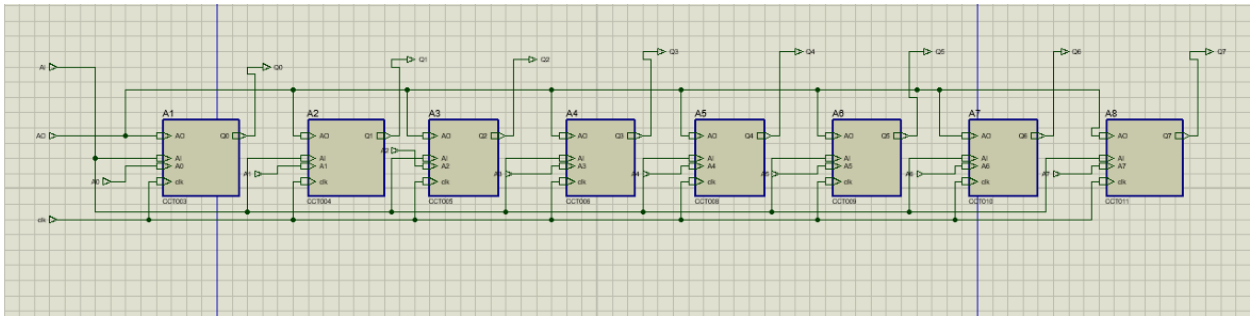


Fig – Accumulator Internal Circuit

Pin	Type	Function
AI	Input	Accumulator In: Stores bus values into Accumulator.
AO	Input	Accumulator Out: Sends stored content to bus.
A0–A7	Input	Takes input from the bus.
CLK	Input	Clock pulse input.
Q0–Q7	Output	Stored accumulator value/result.

2. B-Register

Stores 8-bit value from bus as a fixed operand for the ALU. Does not drive the system bus (no output-enable pin).

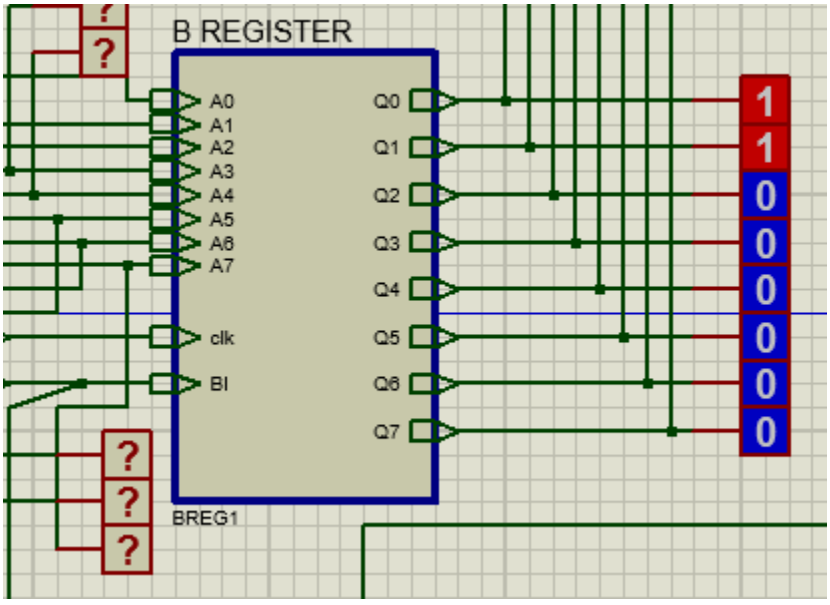


Fig - B-Register Block Diagram

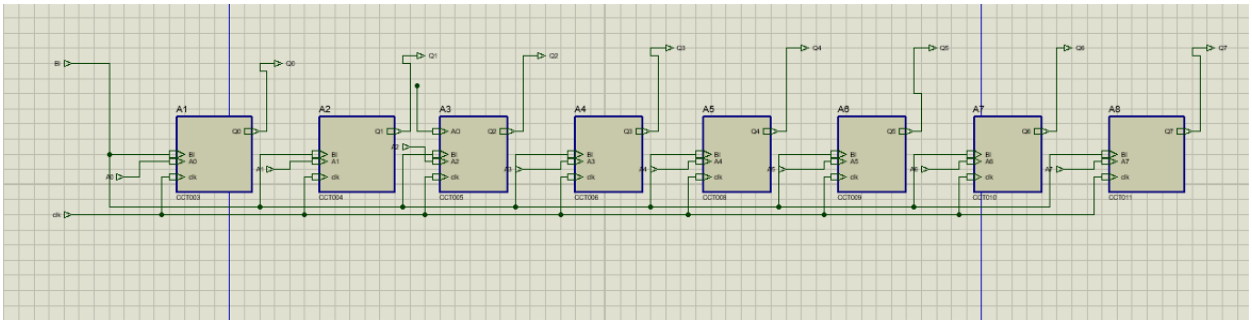


Fig - B-Register Internal Circuit

Pin	Type	Function
BI	Input	B-Register In: Stores bus values into B-Register.
A0–A7	Input	Takes input from the bus.
CLK	Input	Clock pulse input.
Q0–Q7	Output	Stored B-Register value.

3. Memory Address Register (MAR)

4-bit register storing the address of the memory location in RAM to be accessed.

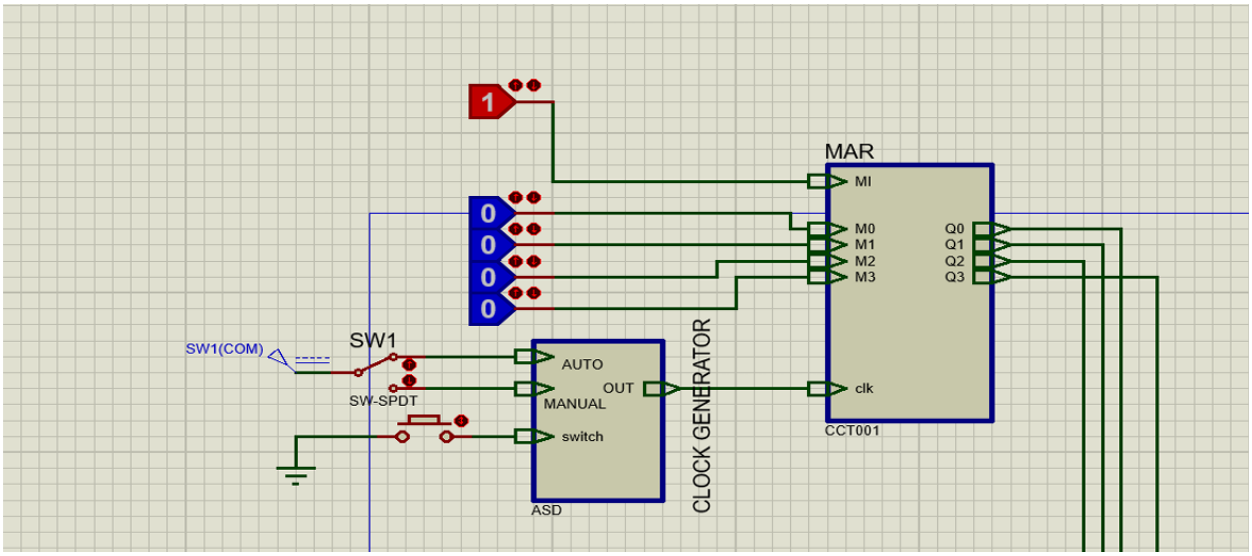


Fig - MAR Block Diagram

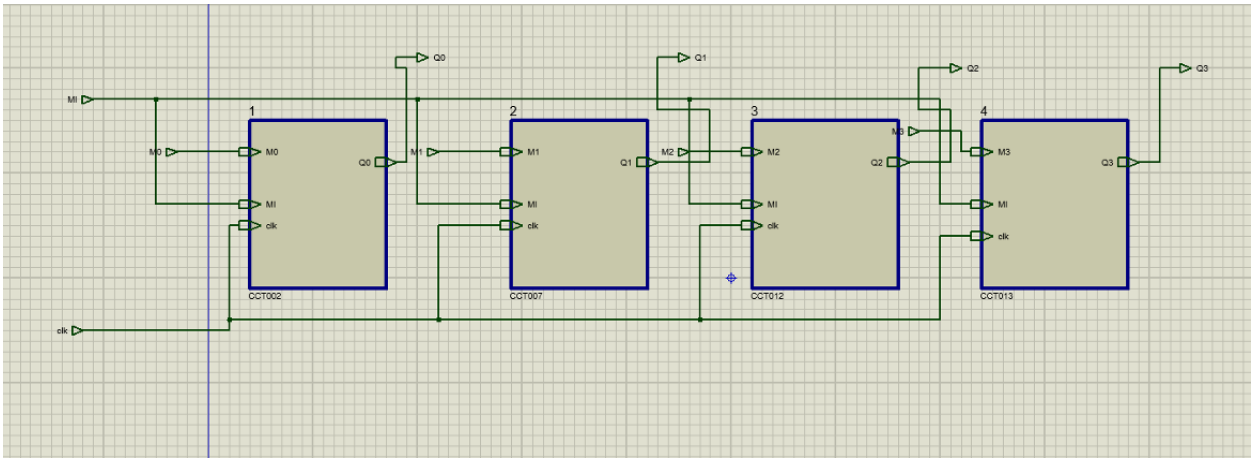


Fig - MAR Internal Circuit

Pin	Type	Function
MI	Input	Memory Address In: Stores bus values into MAR.
M0-M3	Input	Takes input from the bus.
CLK	Input	Clock pulse input.
Q0–Q3	Output	Stored 4-bit address output to RAM.

2.2 RAM System with Input & Selector Unit

Overview

Integrates an 8x16 RAM with an Input Unit and Selector for dual-mode operation (Manual Programming vs. Automated Run).

RAM Components

Uses two 74LS189 ICs (split-bus configuration) for 8-bit data within 16-byte address space.

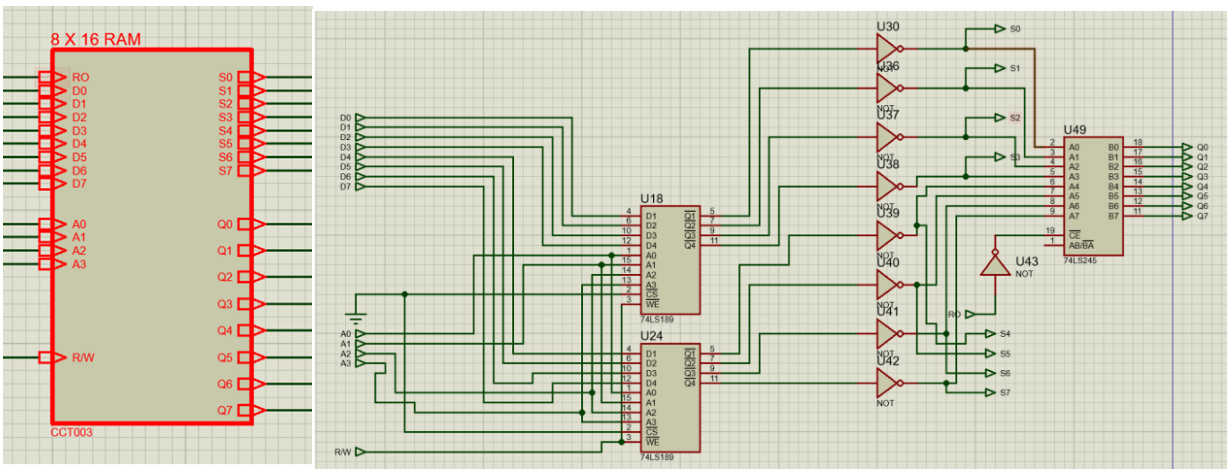


fig – Block and Circuit Diagram of RAM

Pin	Type	Function
RO	Input	Output Enabler: Controls 74LS245 to place RAM Data onto Bus.
D0-D7	Input	Data Inputs from switches.
A0-A3	Input	Address Inputs (Selects 1 of 16 memory slots).
R/W	Input	Read/Write Signal.
Q1-Q7	Output	Bus Data Outputs.
S0-S7	Output	Preview Data Outputs (LEDs).

Input Unit

Provides manual data and address control via physical switches. Controls R/W and Run/Prog states.

Selector Unit

Multiplexer toggling RAM address source between Manual Input (Program Mode) and MAR (Run Mode).

Additional Features

- **Address Display:** Visual reference for current memory location.
- **Preview Signal:** Debugging output for RAM data before bus release.

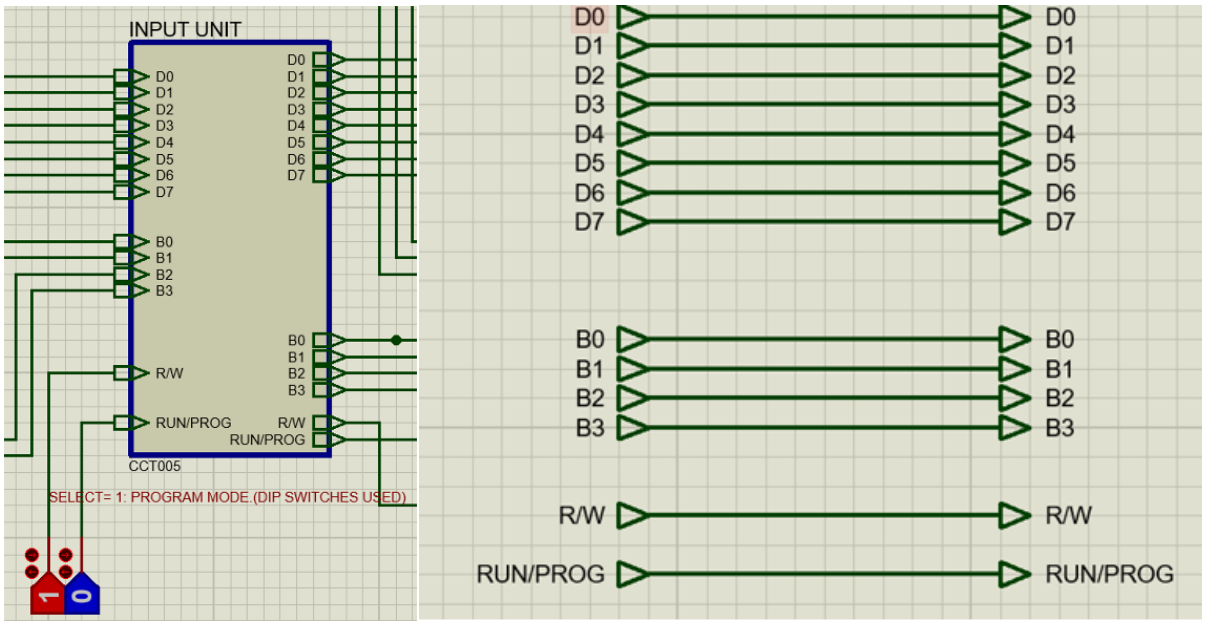


Fig – Block and Circuit Diagram of Input Unit

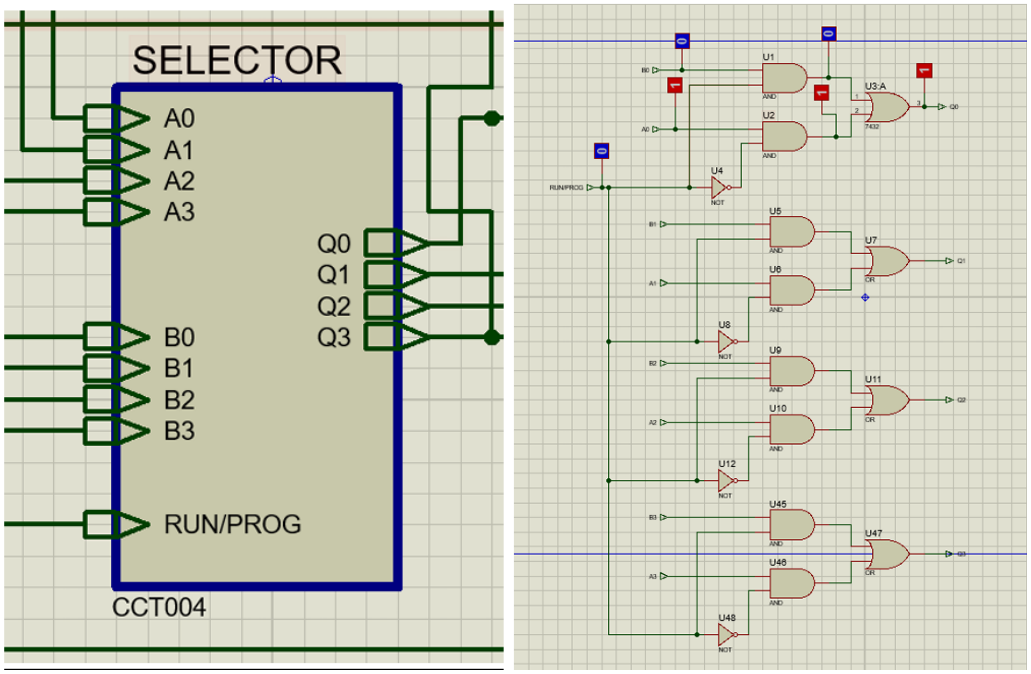
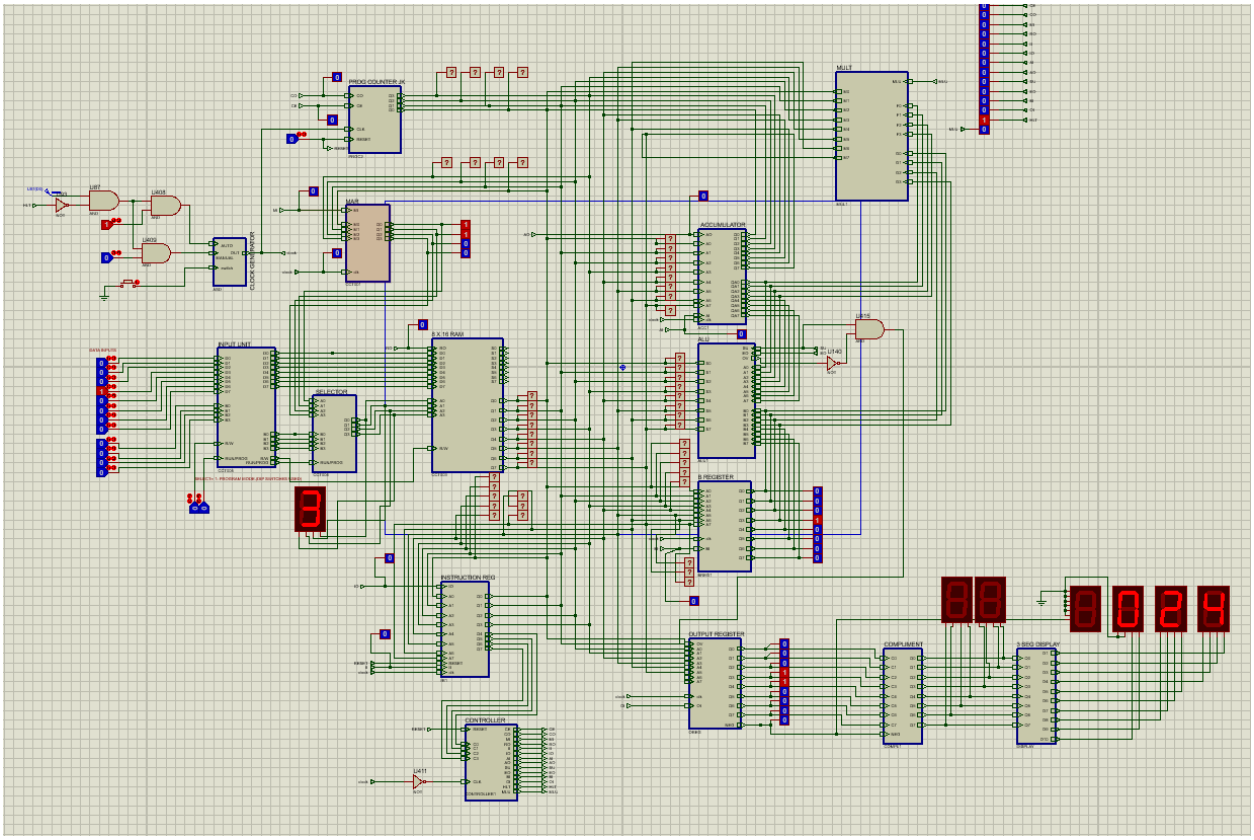


Fig – Block and Circuit Diagram of Selector Unit

Phase C & D: ALU, Controller, Multiplier & Output

Overview

The system is built around an **8-bit data bus** that interconnects all modules. Each module communicates through tri-state logic ensuring only one device drives the bus at a time.



3.1 Arithmetic Logic Unit (ALU)

Components Used

- Two 4-bit adders (4008)
- XOR gates (for subtraction)
- 74LS245 (tri-state buffer)
- Control signals: SU, EO

Working Principle (Refined)

The ALU performs addition and subtraction using a cascaded 8-bit adder:

- **Addition (SU = 0):**
  Direct addition of A and B.

- **Subtraction (SU = 1):**
  Implemented using **2's complement**:
    - XOR gates invert B → 1's complement
    - Carry-in = 1 → forms 2's complement
    - Result = A – B

The output is enabled onto the bus using **EO** via the tri-state buffer.

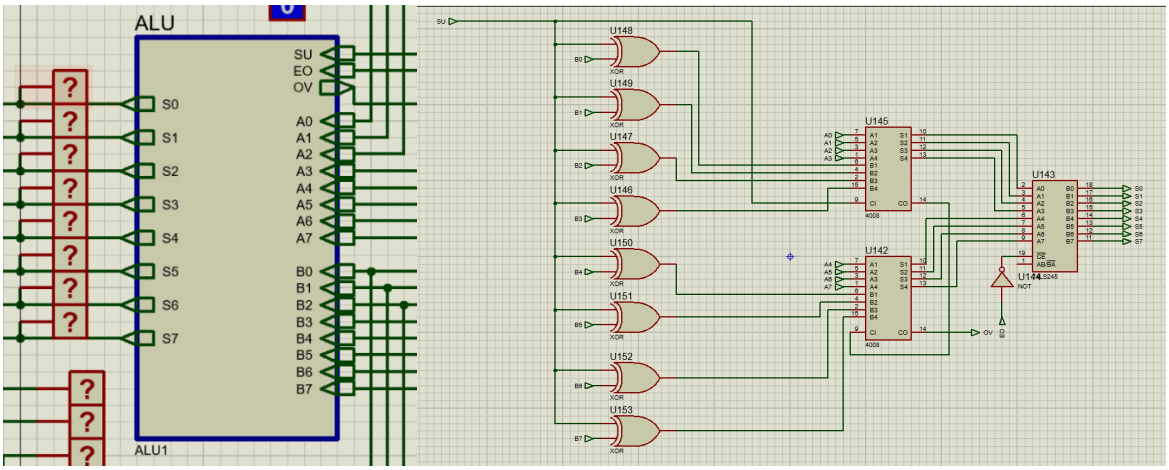


Fig – ALU Circuit

Pin	Type	Function
A0-A7	Input	Takes the 8 bit data from the Accumulator
B0-B7	Input	Takes the 8 bit data from the B Register
SU	Input	When SU=0, addition operation and when SU=1, subtraction operation will be executed.
EO	Input	Puts the ALU output onto the bus.
S0-S7	Output	The 8-bit result of the calculation.
OV	Output	Is HIGH when the subtraction result is negative

3.2 Instruction Register (IR)

Components Used

- D Flip-Flops (4013)
- NOT, AND, OR gates

- Tri-state buffer (74126)

Working Principle

- Stores 8-bit instruction from RAM
- Splits into:
    - **Opcode (Q4–Q7)** → sent to controller
    - **Operand (Q0–Q3)** → sent to MAR

Acts as a buffer ensuring stable instruction execution.

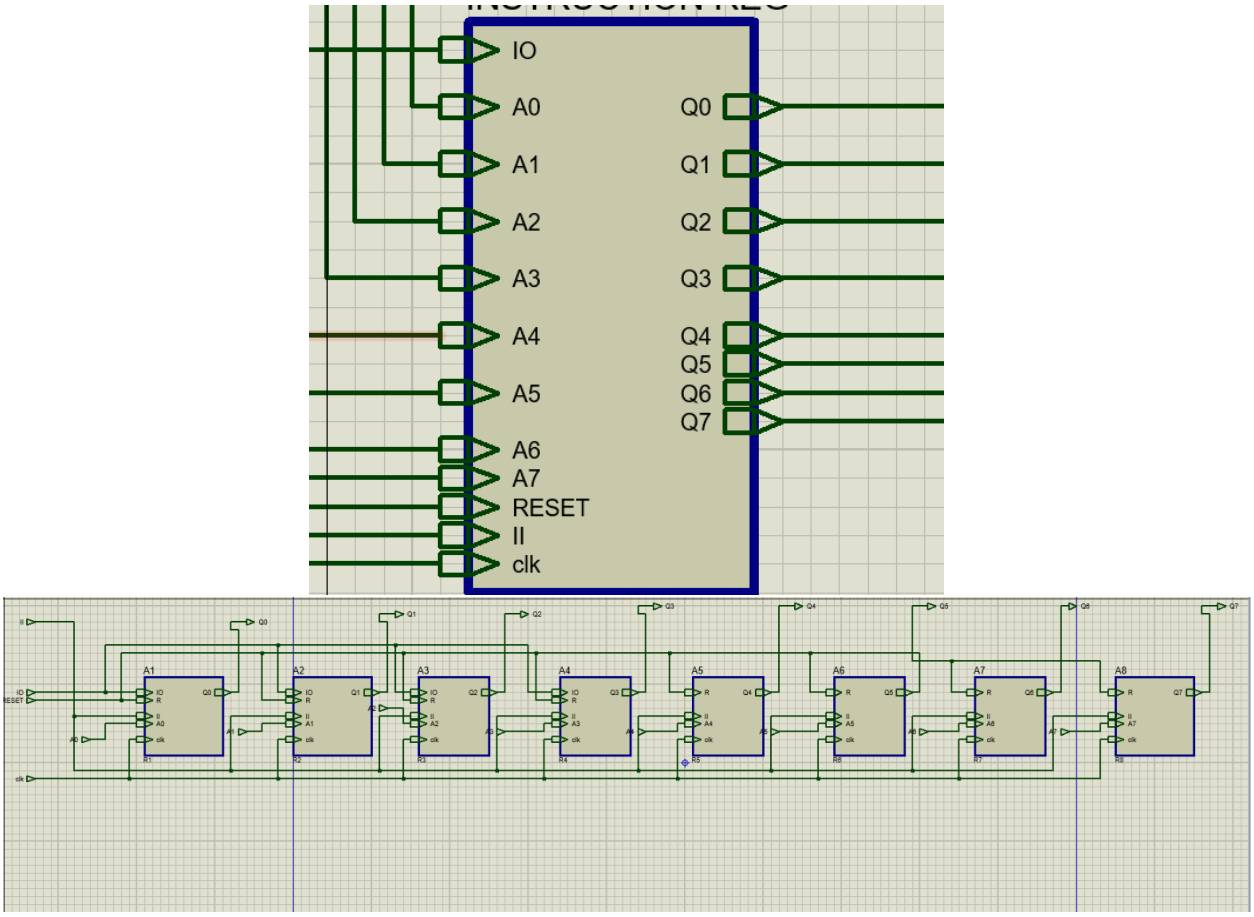


Fig – Instruction Register

Pin	Type	Function
A0-A7	Input	Receive the full 8-bit instruction (OpCode + Operand) sent from the RAM through the bus
II (Instruction In)	Input	Writes the current values of the bus into the instruction register.
clk (Clock)	Input	The clock input from the clock pulse generator

Q0- Q3	Output	Operand Output: Sends address
Q4-Q7	Output	OpCode Output: They are sent directly to the **Controller** to be decoded (e.g., identifying the command as LDA, ADD, or OUT).
IO	Output	Sends the stored data (only lower nibble) onto the bus. Otherwise these bits are disconnected

3.3 Controller Sequencer

Components Used

- D Flip-Flops
- AND, OR, NOT gates

Working Principle

Generates control signals for all modules based on:

- Opcode
- Timing states (T1–T6)

Instruction Cycle

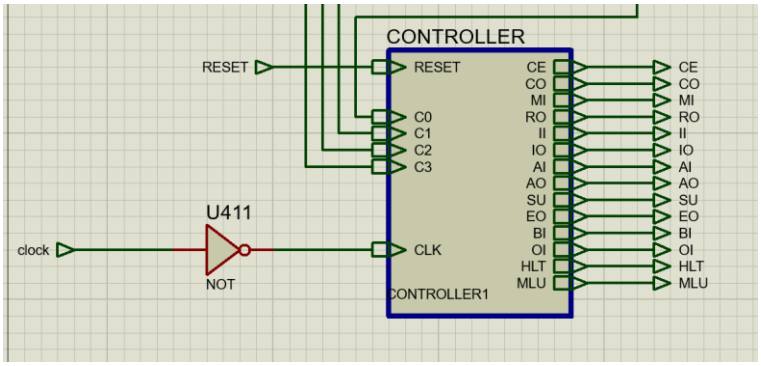
Fetch Cycle (T1–T3)

- T1: PC → MAR
- T2: PC increment
- T3: RAM → IR

Execution Cycle (T4–T6)

Depends on instruction:

- LDA, ADD, SUB, MULT, OUT, HLT



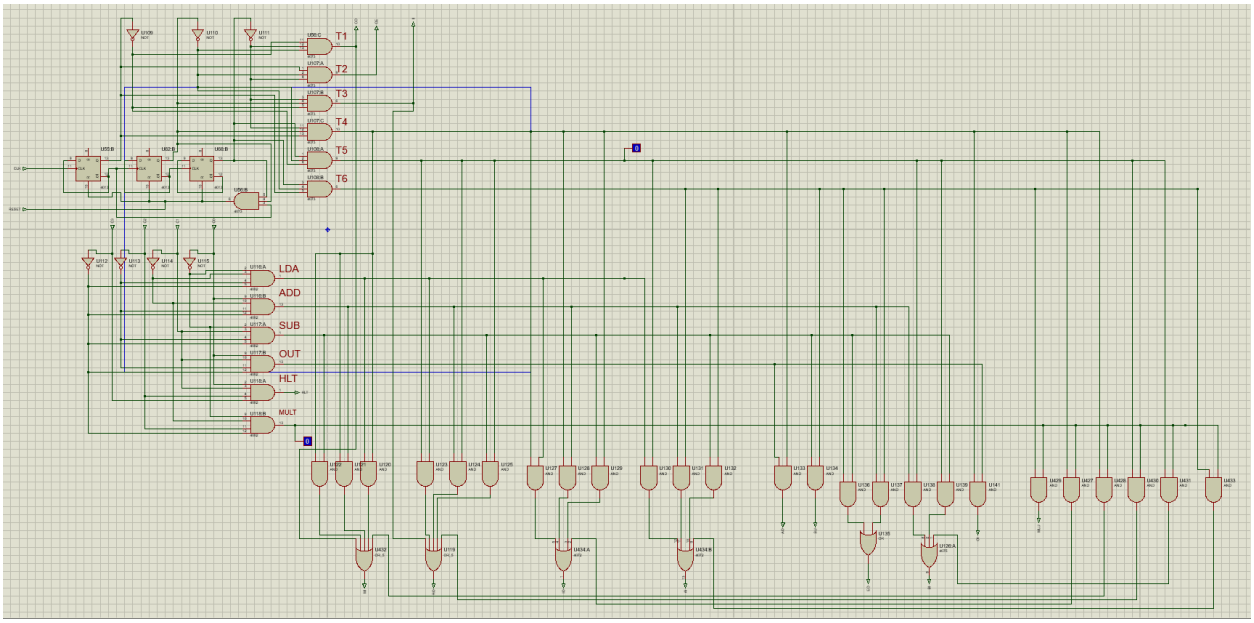


Fig - Controller Sequencer

Pin	Type	Function
C0-C3	Input	Instruction Register opcode inputs. Represent the current instruction opcode
RESET		Resets the internal timing counter (T-state counter) to initial state T1.
CLK		The clock input from the clock pulse generator
CE	Output	Increment program counter
CO		Enables program counter output to bus
MI		Loads MAR from bus
RO		Enables RAM to output data onto bus
II		Loads instruction into IR
IO		Outputs lower 4 bits(Q0-Q3) of IR to bus

AI		Loads data into A register
AO		Places A register content onto bus
SU		Controls ALU operation mode
EO		Enables ALU output onto bus
BI		Loads data into B register
OI		Loads data to output register from bus
HLT		Stops system clock
MLU		Enables the multiplier unit

3.4 Multiplier Unit

Implements a **4-bit binary multiplier** producing an 8-bit result.

Working Principle

1. **Partial Products:** Generated using AND gates
2. **Addition:** Using Half & Full Adders
3. **Output Formation:**
    - M0 → direct
    - M1–M6 → summed outputs
    - M7 → final carry

Control Signal (MLU)

The **MLU signal** is routed through logic (including an inverter and control block). It acts as an enable/ disable control and is linked to the controller unit of our total circuit.

We also added a new controller sequence with the opcode **0100** for multiplication.

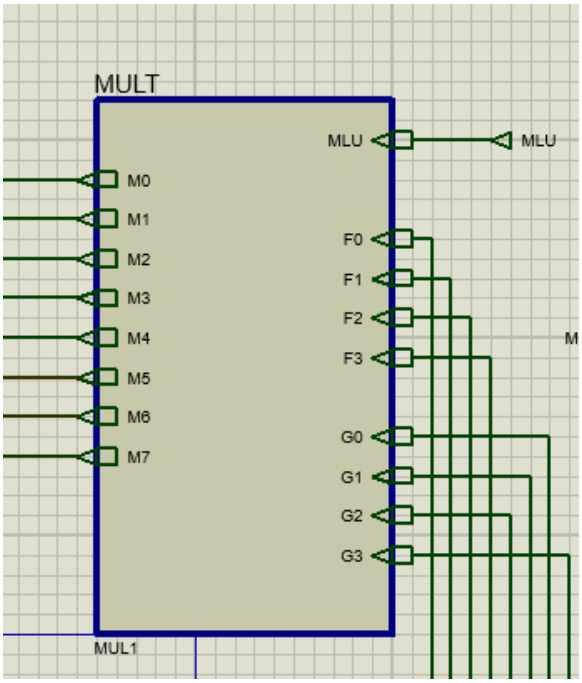


Fig - *Multiplier Outer Circuit*

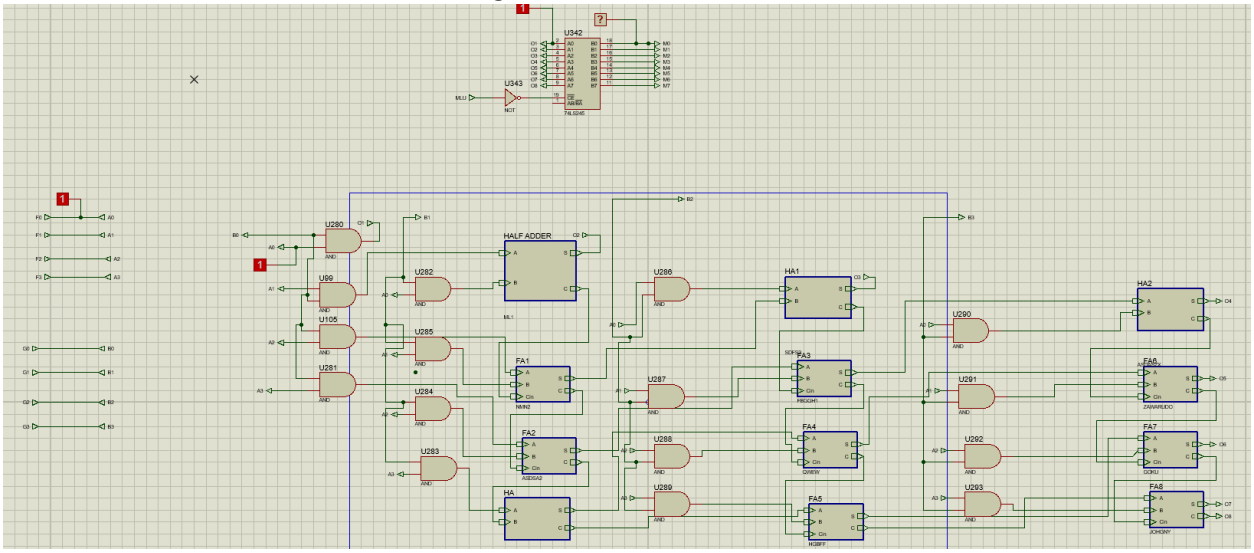


Fig - *Multiplier Internal Circuit*

3.5 Bus System Integration

Working Principle

- 8-bit bus connects all modules
- Only one module outputs at a time
- Tri-state buffers prevent contention

Connected Modules

- PC, MAR, RAM
- IR, ALU, Multiplier
- Registers and Output Unit

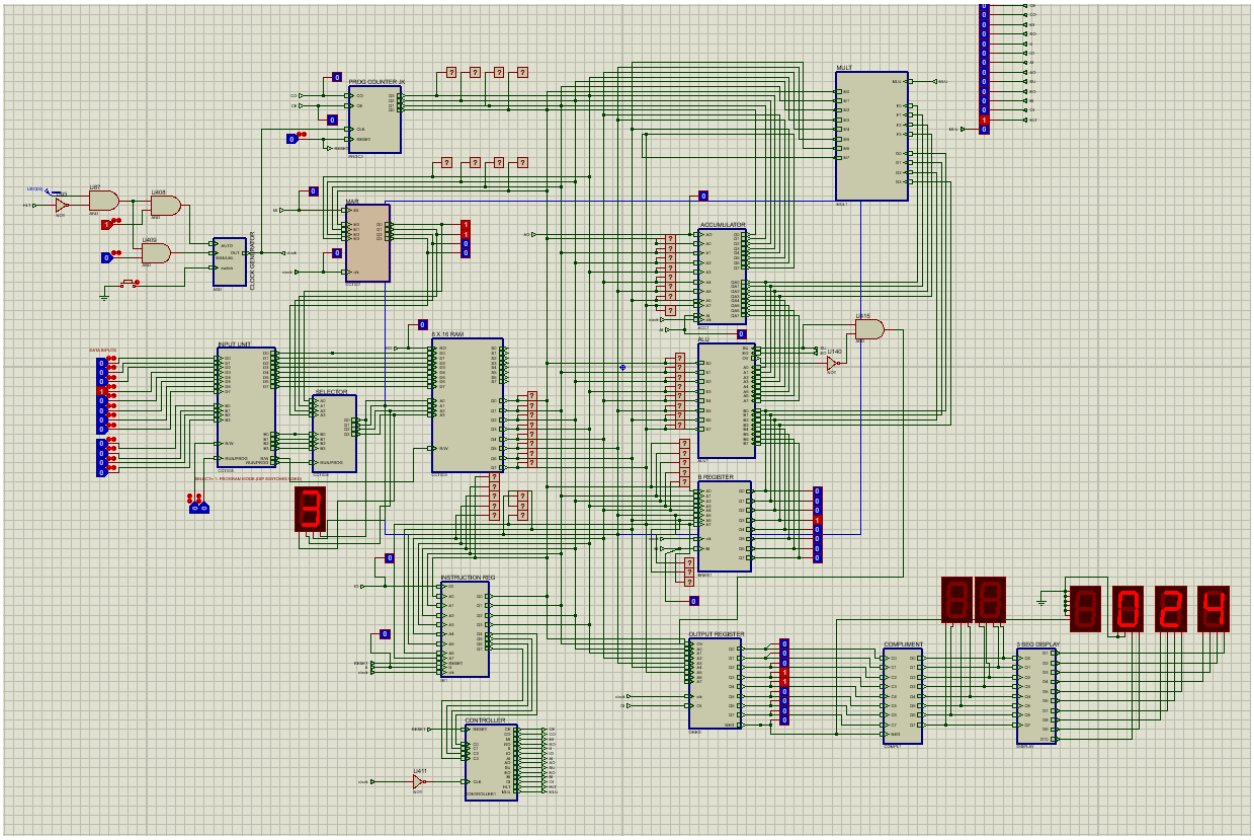


Fig – Bus Integration

3.6 Output Unit

Components

- Output Register
- Display modules (Hex + Decimal)
- Complement Generator

Working Principle

- Loads data using **OI signal**
- Uses **OV flag → NEG signal**
- Negative values:
    - Converted using 2’s complement

- Displayed with negative sign

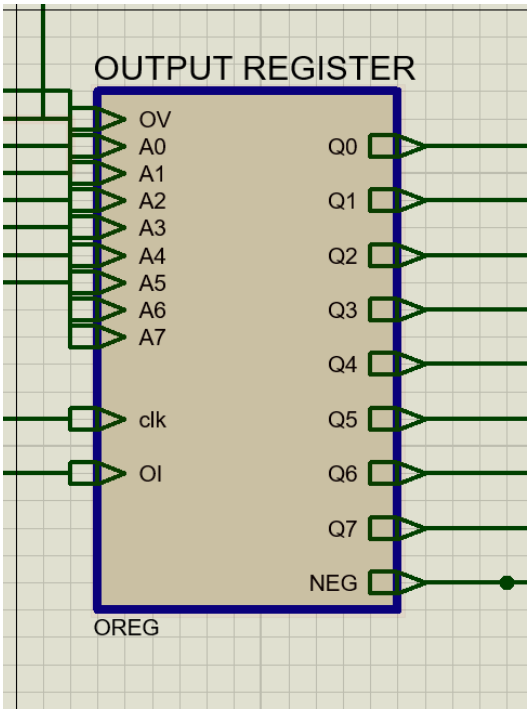


Fig – Outer Circuit of Output Register

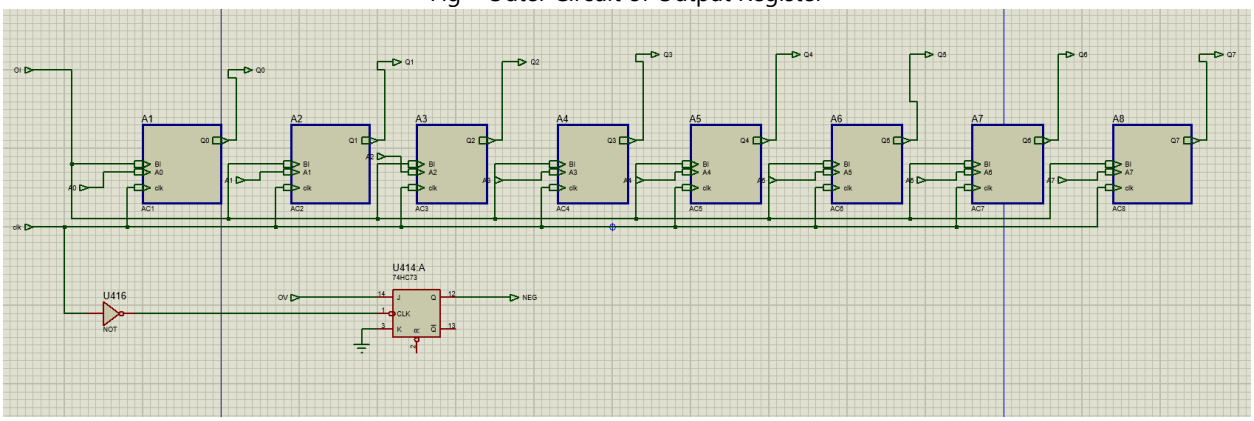


Fig – Inner Circuit of Output Register

Pin	Type	Function
A0-A7	Input	Takes the 8 bit data from the bus
OV	Input	Overflow input from ALU
OI	Input	Controls whether to receive data from the bus or not
Q0-Q7	Output	Stored 8-bit output value to be displayed
NEG	Output	Signal indicating negative result and negative sign display

3.7 BCD Converter for Decimal Display

Objective: Convert 8-bit binary output into readable decimal digits.

Working Principle

- Uses **Shift-and-Add-3 (Double Dabble)** method
- Implemented using cascaded **META blocks**
- Each META block:
    - Adds 3 if input ≥ 5
    - Built using combinational logic

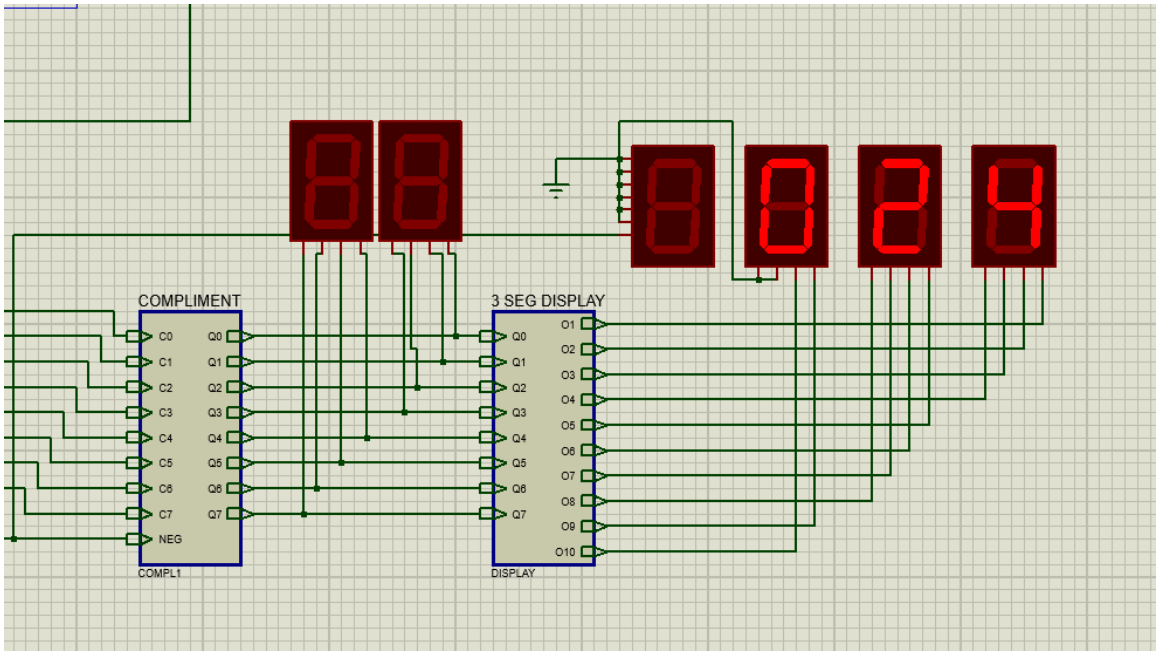


Fig - Top-level BCD converter connection with complement and display blocks.

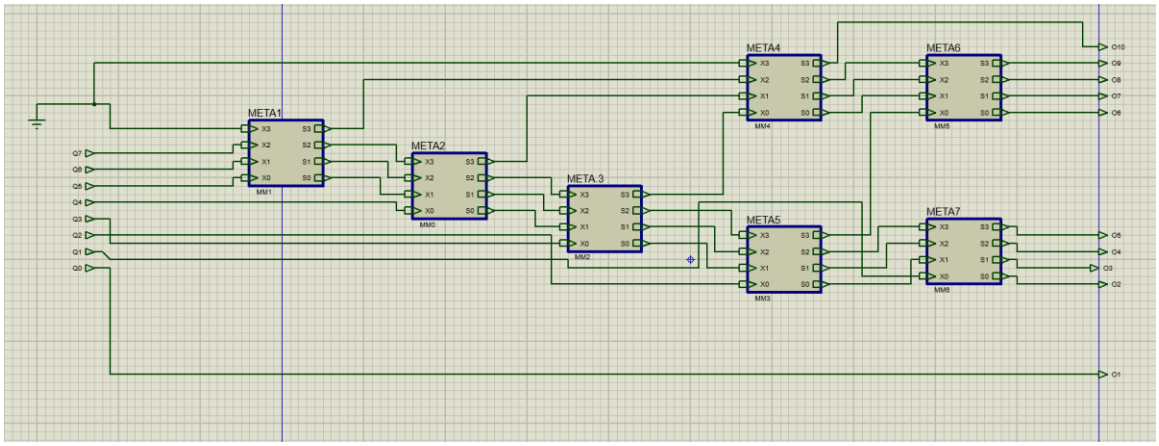


Fig - Internal implementation of the 3 SEG DISPLAY block using cascaded META modules

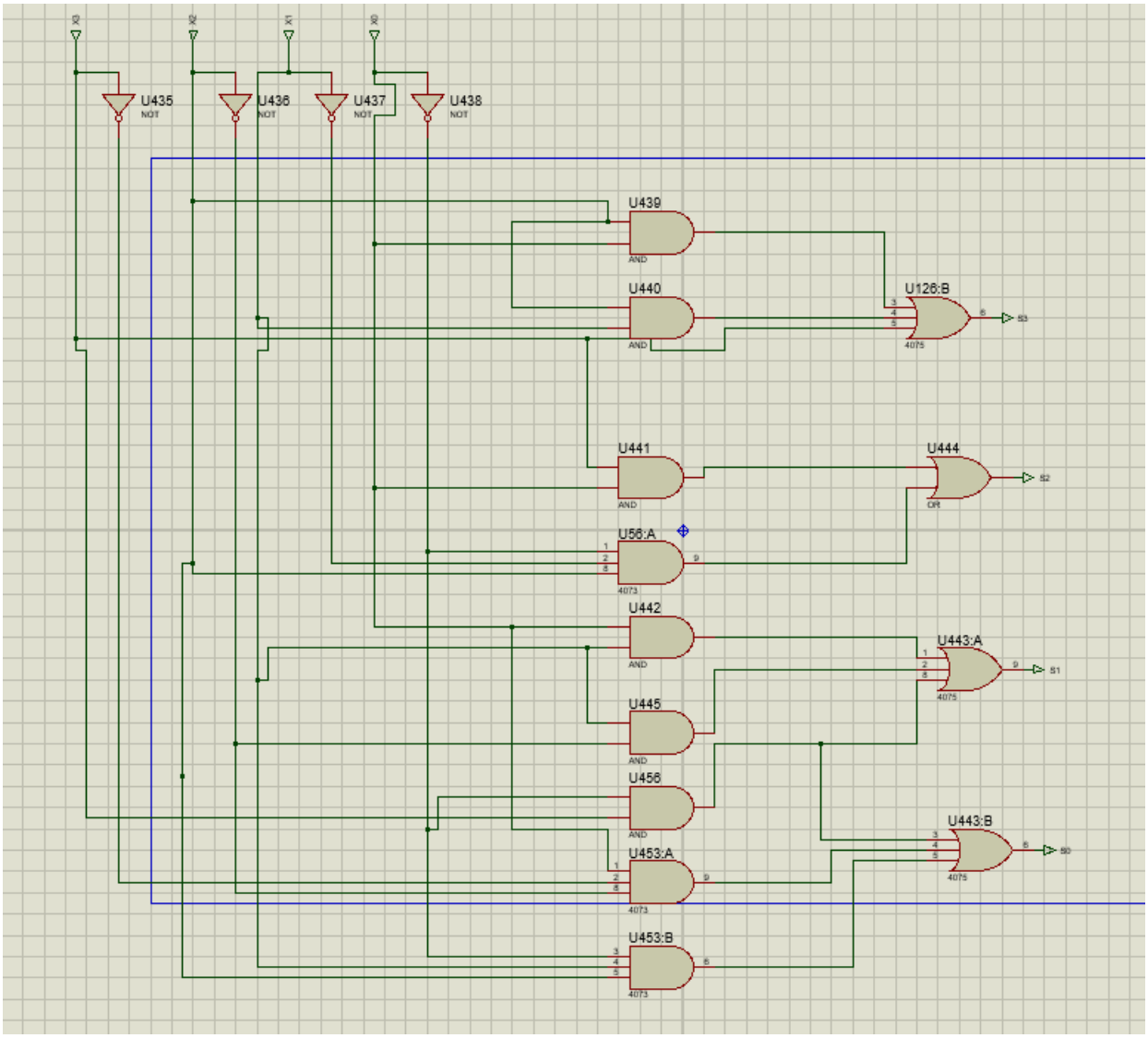


Fig - Gate-level implementation of a META block using NOT, AND, and OR gates.

Key Logic (META Block)

- Performs conditional +3 correction
- Ensures valid BCD output

Truth Table and Boolean Equations of META Block

Each META block adds 3 to the input when the input value is 5 or greater. The truth table is given below.

Input (X3 X2 X1 X0)	Decimal	Output (S3 S2 S1 S0)
0000	0	0000
0001	1	0001
0010	2	0010
0011	3	0011
0100	4	0100
0101	5	1000
0110	6	1001
0111	7	1010
1000	8	1011
1001	9	1100

For inputs 1010 to 1111, the states are not used in normal BCD conversion and may be treated as don't care conditions.

The simplified output equations of the META block are given below. Here, the apostrophe (') denotes complemented input.

$$S_3 = X_3 + X_2X_1 + X_2X_0$$

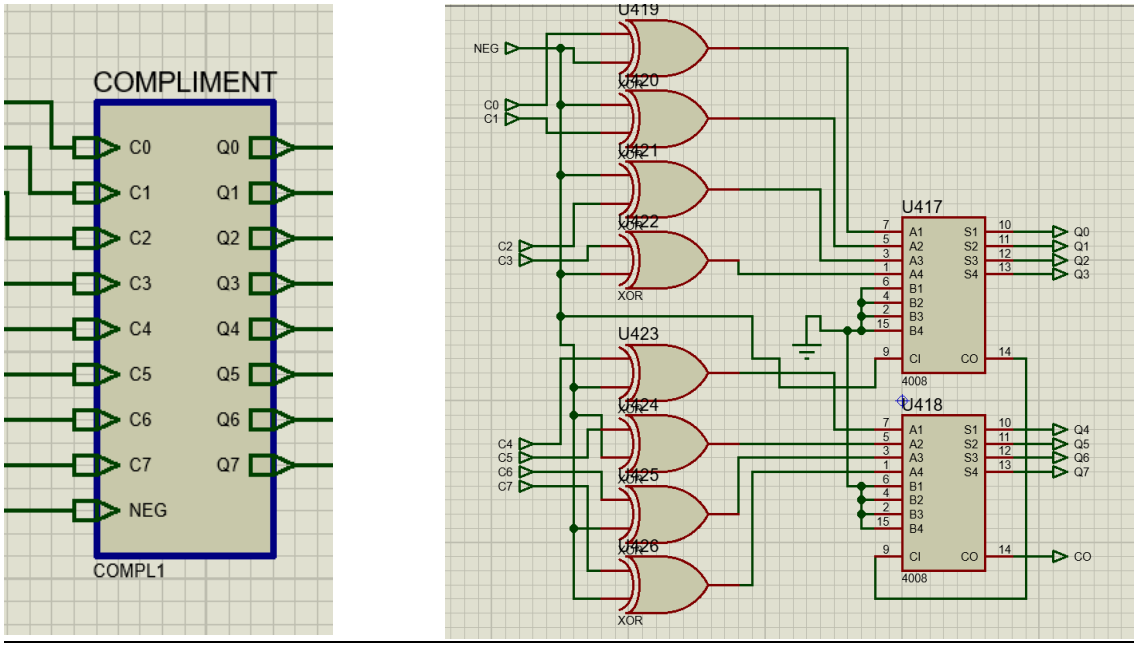
$$S_2 = X_3X_0 + X_2X_1'X_0'$$

$$S_1 = X_1X_0 + X_1X_2' + X_3X_0'$$

$$S_0 = X_3X_0' + X_1X_2X_0' + X_0X_2'X_3'$$

The BCD converter is a useful additional feature of the project. It improves output readability by converting the 8-bit result into decimal digits and displaying them clearly, which makes the overall system more practical and user-friendly.

3.8 Complement Generation module:



This module operates under control of the **NEG signal** (derived from the OV flag) and conditionally transforms the input data before it is displayed. This module ensures that positive values are displayed normally, while negative values would be 2's complemented to produce their magnitude form.

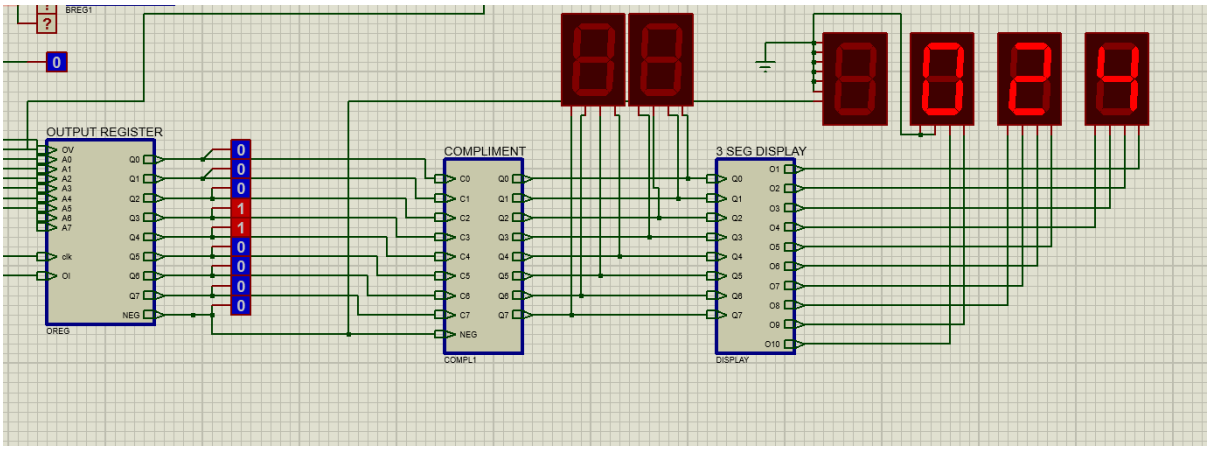


Figure: Full output unit

3.9 Negative sign display:

We added an additional display with the output displays. When the output number is negative and the NEG pin is triggered, the magnitude of the negative number is shown in the regular 7-seg displays and there is another display in front which shows the negative (-) sign in that case.

Features of the System

- 8-bit arithmetic operations
- 4-bit multiplication
- Decimal display using BCD
- Negative number handling
- Modular SAP-1 architecture

12. Conclusion

The system successfully integrates multiple digital modules into a functional SAP-1 architecture. The addition of the multiplier and BCD converter enhances computational capability and output readability, making the system more practical and user-friendly.